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Binary Instrumentation for Tracking Edge Coverage in Fuzzing

Author:

Scandroglio Edoardo

Student ID:

216600

Advisor:

Prof. Name Surname

Academic Year:

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Dedicated to someone.

Abstract

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Keywords: key, words, go, here

Abstract in lingua italiana

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1 | Background and Motivation

This chapter will focus on presenting the context of this thesis. In particular, in Section 1.1 the fundamentals of fuzzing will be discussed, with particular attention to coverage-guided grey-box fuzzing. In Section 1.2 the background on binary rewriting is discussed, along with its approaches and difficulties. Section 1.3 states the problem addressed by this work, and Section 1.4 reviews the state of the art in binary-only fuzzing, also focusing on a description of AFL/AFL++ internals. In the last section (Section 1.5) I will present the main goals and challenges of this thesis, and explain why the implemented tool differs from the state of the art.

1.1. Fuzzing Background

Fuzzing is nowadays a widely used approach for software testing and vulnerability assessment [17, 26]. Its main strength derives from the fact that it can rapidly produce various inputs to be fed into a target executable. In general, a fuzzer can be divided into the **Input Generator**, which is responsible for providing inputs to the **Executor** that executes them. During the execution, the **Defect Monitor** checks whether the executed input causes a crash of the target.

In particular, the Input Generator must choose which input to provide to the executor among all the inputs in the *Input Space*. A subset of the *Input Space* is represented by the *Valid Input Space*. This is a subset of the input space that specifically contains all the inputs that are well formed with respect to the application being tested. For example, if the fuzzed target is a PDF reader, a valid input is a PDF file. A good Input Generator must choose inputs that are also valid inputs, because only by doing this is there a probability of finding a vulnerability [26]. Hence the need for a precise input selection and mutation strategy arises, in order to discover as many edge cases as possible.

In order to do so, the Input Generator must obtain some information from the Executor. Depending on the amount of information the Input Generator has, fuzzing can be classified into *White-box*, *Black-box* or *Grey-box*. In Black-box fuzzing, the fuzzer does not have any knowledge of the execution [19]. In White-box fuzzing, instead, the source

code must always be available, and techniques such as Symbolic and Concolic execution are used [11]. In Grey-box fuzzing, instead, the inputs are chosen using, for example, **Code Coverage** information (i.e. the number of lines of code visited during execution of the target). This approach is called *Coverage-Guided Grey-box fuzzing* and is based on prioritizing the inputs that discover new sections of code. Since the source code is not always available, Code Coverage is not always a viable Coverage Measure. In general, the coverage measures can be of different granularity [10]. The three main options are:

- **Block Coverage:** Keeps track of the hit count for each Basic Block (BB). This approach is used by some popular fuzzers such as libFuzzer or honggfuzz [10]. This method is often too imprecise because it fails to capture the transitions between BBs.
- **Edge Coverage:** Keeps track of the hit count of every edge in the Control Flow Graph (CFG) (i.e. the transition between two BBs). This is the approach used by AFL and AFL++ [8]. It maintains overheads similar to Block Coverage but can track the transitions between BBs and not only their execution, adding some context information. However, it lacks precision in keeping track of the order in which the edges are taken.
- **Path Coverage:** This is the most complete metric and keeps track of the complete path taken during the execution of a specific input, thus also keeping track of the order in which the edges are taken. The main drawback of this metric is that a single execution path could become very large, resulting in large runtime and memory overheads.

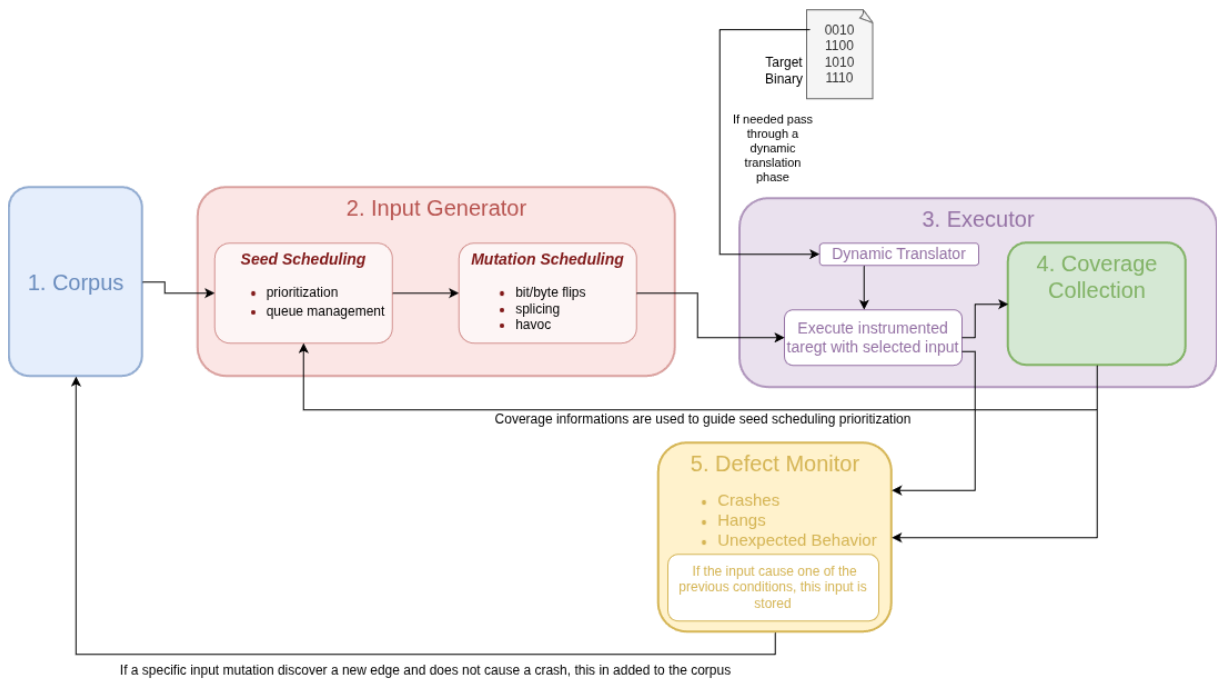
The best approach to Code Coverage, in terms of overhead and effectiveness, is **Edge Coverage** [10], which keeps track of which edges in the CFG are taken during execution without the overheads incurred by keeping track of the entire execution path. Coverage information is obtained through the instrumentation of the target application. Depending on the availability of the source code, the instrumentation can be added in different ways:

- The source code is available: In this case the instrumentation is added at *Compile Time*.
- The source code is not available: In this case there are two different techniques for instrumenting the application:
 - **Static Binary Instrumentation (SBI):** The instrumentation is added before the target is executed, without access to the source code, relying on Static Binary Rewriting techniques.

- **Dynamic Binary Instrumentation (DBI):** The instrumentation is added during program execution, relying on Dynamic Binary Rewriting techniques.

Regardless of the availability of the source code, the *Coverage-Guided Grey-box* fuzzing technique can be summarized as shown in fig. 1.1.

Figure 1.1: Generic diagram describing the Coverage-Guided Grey-box fuzzing workflow



Specifically for **Edge Coverage**, fig. 1.1 can be described as follows:

1. **Seed Corpus:** Represents the set of inputs on which mutations are computed. At the beginning it is composed of user-defined seeds, but during fuzzing this set can grow if a specific mutation discovers new edges not yet found.
2. **Input Generator:** Composed of two main parts:
 - **Seed Scheduling:** responsible for deciding which seed in the corpus to mutate.
 - **Mutation Scheduling:** responsible for deciding which mutation or mutations to apply to the selected seed.
3. **Executor:** The component that actually executes the target binary. If the target binary has already been instrumented with Edge Coverage tracking instructions, it is executed as is; otherwise, in the case of DBI, a dynamic translator must dynamically instrument the binary.

4. **Coverage Collection:** In the case of Edge Coverage, this part is represented by the data structure used to keep track of which edges between BBs have been taken and which have not.

This information is used, in some tools such as AFL++ [8], to guide Seed Scheduling, a technique that, given different inputs with the same level of coverage (i.e. number of edges visited), prioritizes the input that is smaller in size and takes less execution time.

5. **Defect Monitor:** This component checks the exit status of the target to catch any crash or timeout of the target, or any other unexpected behavior. If none of these conditions are met, meaning that the target ends normally, the coverage is checked to see whether the current input discovered a new edge. If the input discovered a new edge, that input is added to the corpus.

1.2. Binary Rewriting Background

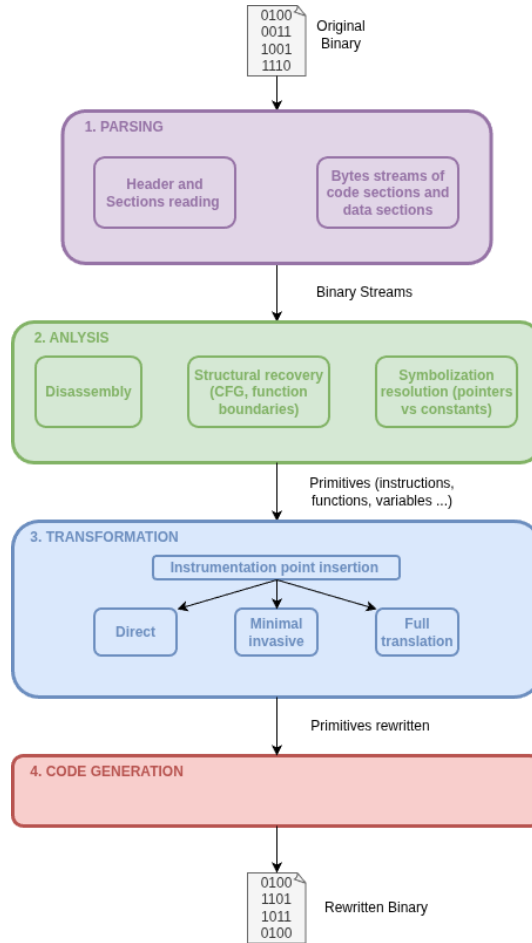
Binary rewriting consists in altering an already compiled executable without having access to the source code, while ensuring that the modified binary remains executable. The process of binary rewriting can be classified into Static Binary Rewriting, if the binary is stored in persistent memory while being rewritten, and Dynamic Binary Rewriting, if the binary is being executed. The process of binary rewriting can be divided into four main steps [22]:

1. *Parsing:* In this phase the main goal is to retrieve information about the binary. In particular, the output of this phase is a raw stream of instructions (in machine code) and data. This information is then used in the Analysis step.
2. *Analysis:* The information gained from the Parsing phase is analyzed to reconstruct the program structure. For example, the raw stream of machine code is passed through a disassembler that reconstructs the assembly instructions. A structural recovery is performed, which includes function detection and CFG reconstruction, and labels and symbols are extracted from the raw stream of data.
3. *Transformation:* Once the CFG has been reconstructed and function and data boundaries have been recovered, the binary rewriter can modify the binary at the Instrumentation Point (IP). These are user-defined locations in the binary where the rewriter must add, modify or delete an instruction. The IP can be coarse- or fine-grained. In the first case, for example, the instrumentation could be inserted at branch level (at the beginning of each BB). In the second case, instead, any

instruction could potentially be modified.

4. *Code Generation*: In this last step, the changes made in the previous steps must be integrated into the binary while preserving its executability. This can be done in different ways. For example, the changes could be inserted into a new section of the executable, redirecting execution to the new code at the IP. Alternatively, this step could also be performed by reassembling a completely new executable starting from the changes made at the IP. This second approach is much harder, because it requires preserving all the references inside the executable and preserving the executable layout.

Figure 1.2: An example of a Static Binary Rewriting Workflow

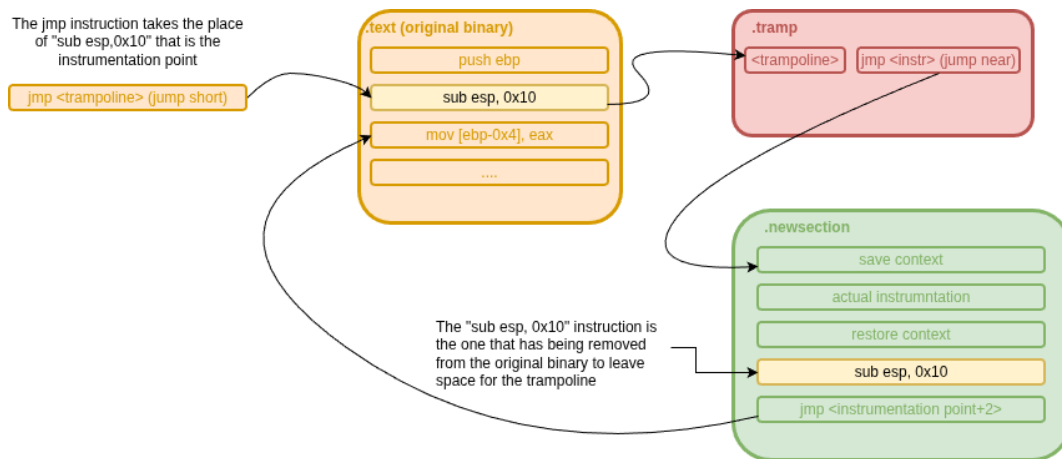


In fig. 1.2, a general scheme for static binary rewriting can be seen. In this case all the phases occur prior to execution, so the cost in terms of overhead is paid only once, before the execution. The main problems are disambiguating between code and data (i.e. deciding whether a stream of bytes represents machine code instructions or constants placed inline in the .text section) and resolving indirect branches [3, 22]. This problem

is generally undecidable and can only be solved using heuristics. There are three main families of static transformation:

- **Direct:** Consists in directly modifying the instruction. Works well with RISC systems with instructions having the same length. Generally fails when adding or removing instructions, because this does not preserve the offset into the `.text` section.
- **Minimal-invasive:** The new code is inserted into a new section of the binary, and at the IP an unconditional jump to the new section is placed. The main limitation of this approach arises when the new section is far from the IP and a near jump (necessary to reach the new section) uses a different number of bytes than the instruction at the instrumentation point. In fact, in x86-64 assembly a short jump uses 2 bytes while a near jump uses up to 5 bytes. So, if a short jump is needed at the instrumentation point for alignment reasons, but the new section is too far away and a near jump is needed instead, it is necessary to add a **trampoline**. This approach can be seen in fig. 1.3. Trampoline-based instrumentation is one of the most common approaches to binary rewriting [22].
- **Full-translation:** The binary is lifted into an Intermediate Representation (IR) that makes it possible to add an IP at instruction level. The IR is then reassembled into machine code after the changes are made.

Figure 1.3: An example of Static Instrumentation using the trampoline technique



With regard to Dynamic Binary Rewriting, the Analysis and Transformation phases are executed at runtime for each instruction. In this case, none of the problems that arise in Static Binary Rewriting occur, since the binary is being executed, so indirect branches can be resolved using the values in the registers, and there is no need to disambiguate between constants inside the `.text` section and instructions. The main drawback of this approach

is the prohibitive overhead caused by the dynamic analysis and transformation of each instruction, together with the fact that only the taken branches can be modified. In fact, if a branch is not actually taken during execution, its instructions never reach the rewriter.

A specific type of binary rewriting is Binary Instrumentation. Binary instrumentation uses binary rewriting techniques to add instructions that track binary behaviour, log events, or, in the case of this thesis, insert edge coverage tracing instructions.

1.3. Problem Statement

Coverage-Guided Grey-box fuzzing is one of the most effective techniques for automatic vulnerability discovery in software [6, 20], due to its ability to generate a wide range of inputs based on the feedback obtained through Code Coverage information, which guides the fuzzer towards unexplored regions of code and therefore helps it find hidden vulnerabilities. In fact, *Black-box* fuzzing lacks a precise mutation strategy based on execution monitoring, while *White-box* fuzzing incurs high overheads due to Symbolic or Concolic execution. Furthermore, the latter approach also requires the availability of the source code, which is not always available. So *Grey-box* fuzzing seems to be the most promising trade-off between precision of input mutation and overhead. However, *Grey-box* fuzzing also has its own drawbacks. As a matter of fact, the effectiveness of this approach strongly depends on the availability of the source code. In fact, both SBI and DBI have many limitations. The main limitation of DBI is the huge amount of overhead (higher than 600% in the case of AFL++ in QEMU mode [20]), which makes the exploration of a substantial part of the valid input space almost impossible. Instead, the main difficulty of SBI arises in modifying already compiled binaries. There are several problems regarding Binary Rewriting, such as Indirect Control Transfers (e.g. `call rax`), shifts in binary sections, or instructions that require rewriting the instructions that reference these sections, and many others, as seen in the previous Section 1.2.

For all these reasons, the preferred technique nowadays is *Compile-time Edge Coverage instrumentation* with fuzzers such as AFL++, LibAFL [9], or libFuzzer [15]. However, in practice, for reasons of industrial secrecy (i.e. proprietary software) or due to precompiled libraries, applications are often closed source, and so Compile-time Edge Coverage instrumentation becomes unusable.

In these scenarios, binary-only fuzzing is the only viable approach. Although SBI presents many challenges, it nonetheless proves to be the most fruitful area, given its low overheads [20]. In this thesis I will address the problem of statically instrumenting a binary with Edge Coverage instructions in a way that makes the instrumented binary compatible with

AFL++, which has been chosen as the fuzzer due to its popularity and because it represents the current state of the art, implementing most of the recently discovered techniques demonstrated by Andrea Fioraldi et al. [8].

The instrumentation must add as little overhead as possible with respect to compile-time overheads and must preserve the correctness of the binary. This is important in fuzzing, because lower overhead means more executions per second, so a larger portion of the valid input space can be explored.

1.4. State of the Art

In this section I will present some of the current state-of-the-art solutions in binary-only grey-box fuzzing, distinguishing between SBI and DBI approaches.

- DBI

- **AFL++ in QEMU/FRIDA mode** [8]: AFL++ in QEMU mode uses QEMU in *User-mode Emulation*, which consists in translating guest machine code (i.e. guest BBs) into an intermediate representation called Translation Block (TB) and subsequently appending the coverage tracing instructions at the end of it. This approach is based on emulating the entire CPU using QEMU’s virtualized environment. As stated before, AFL in QEMU mode has overheads on the order of 600% with respect to compile-time instrumentation [20].

AFL++ in FRIDA mode [1], instead, is based on shared library injection, in which a Frida [21] component called **Stalker** intercepts any BB that is about to be executed, copies it into a **Code Cache** and dynamically instruments the copy with coverage tracing instructions. Execution control is then transferred to the copy, which is executed and then re-intercepted by **Stalker** at the end. Unlike QEMU mode, it does not emulate the CPU, but instead executes within the memory space of the target program. AFL++ reports that FRIDA mode is approximately 2 to 5 times slower than compile-time instrumentation [2].

- **DynamoRIO/Pin** [4, 16]: Both these approaches share the same basic principle, called **Just in-time (JIT) code caching**. This consists in taking control of the thread when the target application starts, and from that moment on, every BB is intercepted before it is executed and is copied into a private **Code Cache**. The instrumentation is then added to the copy, and control is transferred to the code cache. The instrumentation process is more or less identical to FRIDA’s Stalker mechanism, but is not as optimized. Also in this case, the

implemented version uses AFL [25] as the fuzzer [13, 14].

- **SBI**

- **Retrofit** [7]: This tool performs symbolic lifting on the code, replacing every reference discovered during disassembly with a symbolic label. Doing this allows instructions to be added without modifying the references, achieving reassembleable assembly. While Retrofit can add ASan instrumentation, it cannot natively add AFL-style edge coverage instructions. To do so, the lifted assembly must be instrumented with the afl-gcc tool in order to produce the patched binary. Moreover, community-reported issues [18] document missing instrumentation points when Retrofit’s symbolized output is compiled through afl-gcc, caused by a mismatch between Retrofit’s label-naming convention and afl-as’s basic-block detection heuristics. Finally, Retrofit only works with PIE, non-stripped binaries.
- **AFL-Dyninst** [12]: AFL-Dyninst employs Dyninst’s binary rewriting capabilities to instrument an existing executable by inserting callbacks at selected basic blocks. Dyninst’s instrumentation mechanism is based on code patching and trampoline redirection [5], which causes many context switches and state saving/restoring operations. Its performance is slightly better than DBI, but this approach still reaches 500% fuzzing overhead, as demonstrated by Stefan Nagy et al. [20].
- **ZAFL** [20]: This tool uses two main phases: the first consists in reconstructing an IR using the Zipr binary rewriter. The resulting IR is then used by ZAX to perform an optimization on the IR, find the instrumentation points, and finally instrument these points. ZAFL works by inserting calls to a function that updates the shared map at the beginning of each BB. Despite ZAFL’s many improvements, Jifan Xiao et al. [23] demonstrated that it still causes crashes in 28% of the binaries used in their cited paper.

Before proceeding to the next section, in which I will discuss the main goals of this thesis, I want to discuss the core functioning of AFL and AFL++, given its importance for understanding my approach.

1.4.1. AFL Functioning

[24] As previously stated, AFL and, in particular, AFL++ are among the most widely used grey-box fuzzing frameworks [8]. AFL++ represents the community-based continu-

ation of the original AFL project. The core functioning is virtually the same and consists of a Shared Memory (SHM) region shared between the fuzzer and the target. The SHM size can vary, but the default is 65536 bytes.

Each “cell” of the SHM contains the number of times a given edge in the CFG has been taken during the execution of the target. Since the SHM has 65536 “cells”, each “cell” can contain a number from 0 to 255. The “cell” position is computed with a specific hash function applied to the IDs of the two BBs that form the edge. The main Edge Tracing update rule is summarized in algorithm 1.1. Line line 3 of algorithm 1.1 is used to avoid

Algorithm 1.1 AFL-style SHM update rule

```

1:  $idx \leftarrow current\_bb\_id \oplus previous\_bb\_id$ 
2:  $shared\_map[idx] \leftarrow shared\_map[idx] + 1$ 
3:  $previous\_bb\_id \leftarrow current\_bb\_id \gg 1$ 

```

updating the same SHM location when an edge is traversed forward or backward. An example of this could be a simple **do-while** loop, as seen in algorithm 1.2 and fig. 1.4.

The main idea behind AFL is to instrument each basic block of a target application with

Algorithm 1.2 Do-While Example

```

repeat
  {Basic Block A}
until Permanence_Condition
  {Basic Block C}

```

Without line 3 of algorithm 1.1, the hash of Edge1 would result in $ID_A \oplus ID_B$, and the hash of Edge2 (if the permanence condition is not met) would result in $ID_B \oplus ID_A = ID_A \oplus ID_B$, due to the symmetry of XOR. But since the hash is the same, there would be a hash collision that would cause the same “cell” to be updated twice instead of once.

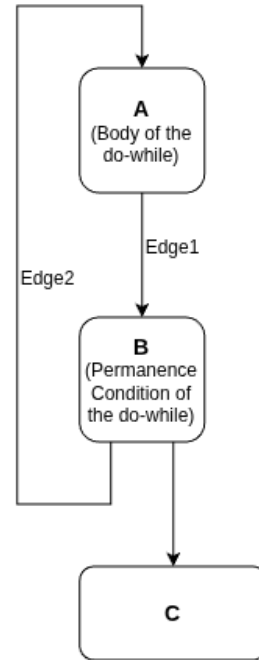


Figure 1.4: Do-while CFG

the shared map update rule. The fuzzer initializes the SHM memory with zeros, runs the instrumented target, which updates the SHM according to algorithm 1.1 until it reaches the end, and then returns to the fuzzer. The SHM ID is communicated to the target through an environment variable called `__AFL_SHM_ID`.

At this point the fuzzer checks the SHM it has just obtained, called `trace_bits`, searching for any newly discovered edge. This is done by keeping a structure called `virgin_bits`, which contains all the edges not yet discovered and is initialized with ones. The comparison is performed through `trace_bits & virgin_bits`. If any position results in one after the logical AND, this means a new edge has been discovered. In this case, the `virgin_bits` data structure is updated and the input just executed is saved in the corpus. Otherwise, a new input is chosen, mutated and then executed.

This approach requires a `fork()` call followed by an `execve()` call for each input, causing significant overhead for each input execution. Two main optimizations have been implemented:

- **forkserver**
- **persistent mode:** (available only in AFL++)

Forkserver

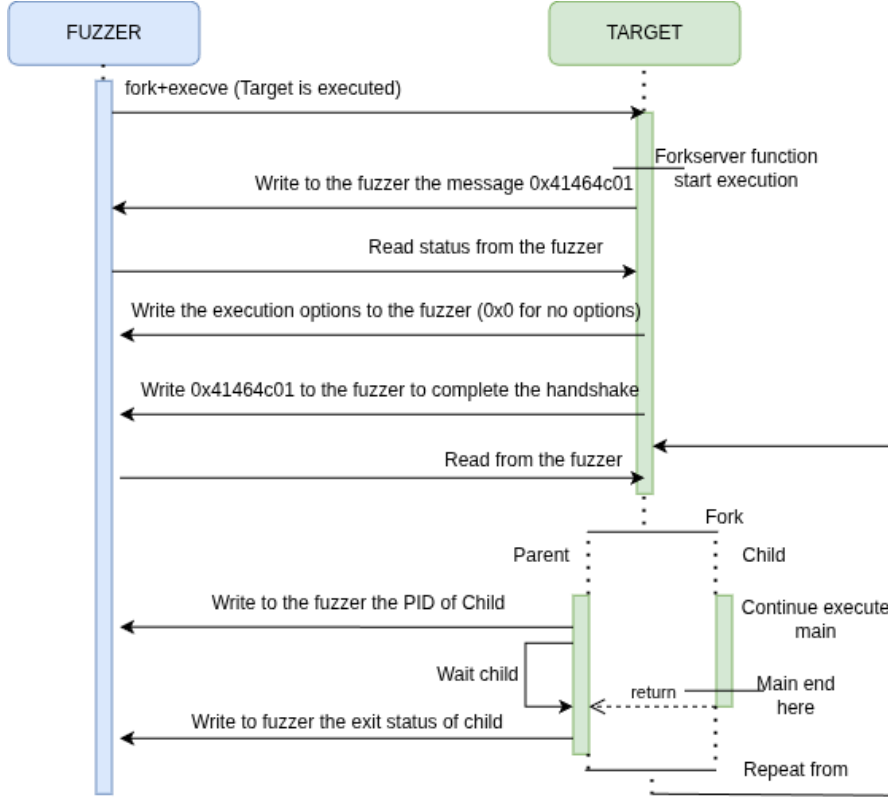
The forkserver makes it possible to avoid calling `execve` for every input, performing it only once at the start of execution. The forkserver is instrumented before the main function of the target application. The AFL++ forkserver protocol can be summarized as follows: ¹ The communication between the Fuzzer and the Target shown in fig. 1.5 occurs unidirectionally through two different file descriptors created by the fuzzer:

- `FORKSRV_FD` (*fd 198*): used by the forkserver to read a message sent by the fuzzer.
- `FORKSRV_FD + 1` (*fd 199*): used by the forkserver to write a message to the fuzzer.

1. The forkserver sends the “Welcome” message to the fuzzer (0x41464c01).
2. The fuzzer replies with the status code. At this point the forkserver must check whether `status & 0xffffffff = 0x41464c01`. If it is equal, execution continues; otherwise, the forkserver stops here.
3. The forkserver sends the execution options to the fuzzer. These can indicate that the target is running in persistent mode, or other optimizations.
4. The forkserver sends 0x41464c01 to the fuzzer to end the handshake.

¹The differences between the AFL forkserver protocol and the AFL++ forkserver protocol lie mainly in the messages exchanged between the forkserver and the fuzzer

Figure 1.5: AFL++ Forkserver Protocol



5. The fuzzer writes a message to the forkserver (its content is discarded); the handshake is now finished. This message is the “go” signal for the forkserver.
6. The forkserver invokes `fork()`. The parent writes the PID of the child to the fuzzer and then invokes `waitpid()`.
The child continues execution of the main function until it ends.
7. When the parent returns from the `waitpid()` function (i.e. the child has terminated), the exit status of the child is sent to the fuzzer, and the forkserver waits for another “go” from the fuzzer.

1.5. Goals and Challenges

The main goal of this thesis is to develop a tool capable of inline, static edge-coverage instrumentation that preserves binary correctness while achieving performance comparable to compile-time instrumentation.

I have chosen to implement the SBI approach instead of DBI due to its low overhead. To do this, I will use the GRAPE binary rewriting framework, a novel approach for binary rewriting that preserves binary executability while introducing small overheads compared

to a trampoline-based approach. This approach involves several challenges, mainly related to preserving the binary's correctness and behavior. Moreover, the tool must be compatible with the already existing AFL++ fuzzer. This means implementing the forkserver communication protocol inside the target binary without recompiling it. This has created no small number of problems, mainly regarding the preservation of the ELF layout after adding new sections. Compared to the ZAFL approach, the SHM update instructions are placed directly inside the BB, rather than in a function called at each BB. This approach drastically increases the size of the file, causing many instruction repetitions (since the SHM instructions are always the same), but was chosen because it does not cause a context switch at each BB. In fact, in order to call a function, the state of the registers must be saved and restored. Moreover, the stack must be allocated, and the callee's RIP must be saved on the stack alongside the old base pointer of the stack. Performing all these operations at each BB would introduce significant overhead.

2 | Approach

In this chapter I will disclose the general aspects of the approach regarding the insertion of the Edge Tracing instructions inside an already compiled binary.

There are three main steps to be fulfilled to have a binary that is able to trace Edge Coverage and to be compatible with AFL/AFL++:

1. Insertion of the forkingserver invocation prior the main (section 2.1).
2. Search of instrumentation points to trace Edge Coverage.
3. Instrument the previously found points with Edge Coverage tracing instructions.

2.1. Forkserver Insertion

This has been one of the most crucial points of this work. First of all I have tried to instrument the forkingserver function directly inline. This approach present many problems. The biggest difficulty is the fact that a forkingserver needs some library functions to work properly :

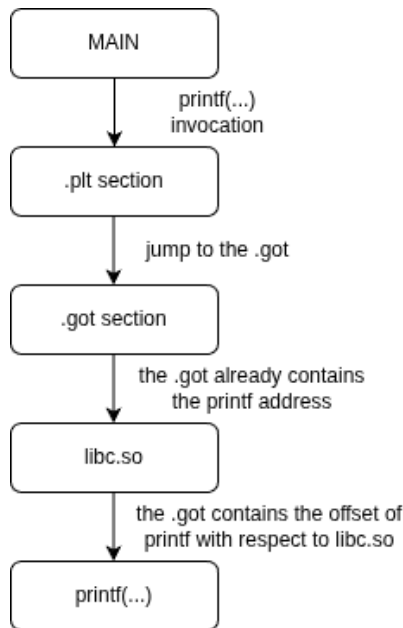
- `read/write` to implement the communication protocol with the fuzzer.
- `fork/waitpid` to implement the forkingserver main logic.

These functions could be part of the target binary (if the binary already uses these functions) or not. In the best case the binary already uses these functions and they are available. In this case a call to these functions could be instrumented at any time taking into consideration the type of linking with which the binary has been compiled.

- In case of *Static Linking* the function instructions are already inside the target application, because during the compilation the linker inserted it. So a call to these functions could be inserted finding the function address and then instrument a call instruction to that address.
- In case of *Dynamic Linking* the function instructions are not inside the target application when the application is launched, but the function addresses are dynamically

resolved by the dynamic linker at need. In this case two new section must be inserted (.plt and .got sections). As can be seen in fig. 2.1

Figure 2.1: An example of printf function resolution using Dynamic Linking and Immediate Binding



In fig. 2.1 can be seen an example of dynamic resolution of `printf` function. In case of Dynamic Linking the `printf` function call in the main is not done to the actual `printf` libc function. Instead there is a specific function in the .plt section usually called `printf@plt`. Into the .plt section each library function is replaced by the following lines of code

```

1  jmp     QWORD PTR [rip +
2      offset_to_printf@GOT]
3  push    0x0
   jmp     <plt_starting_address>

```

Into the .got section there are the resolved addresses of the external functions. If the .got actually contains the resolved address of the `printf` (the library address), the first instruction will jump into the library. The second and third instructions are used in case of **Lazy Binding**. In this case the GOT entry does not contain the address of the library function when the main start, but contain the next address of the plt entry. At the `plt_starting_address` there is the plt resolver that invoke the dynamic linker that search for the address of `printf` function into `libc.so` library.

If the functions required by the forkserver are part of the target binary, their calls can be easily instrumented. Depending on whether the binary is statically or dynamically linked, this can be done either by finding the address of the function and calling it directly, or by finding the address of its .plt entry and calling it, leaving the dynamic linker to resolve the function. The problem arises when the original binary does not include these functions. In this case, it is almost impossible to modify the .plt and .got sections to include the new functions while preserving the binary layout and correctness in the general case. In fact, in some exceptional cases, padding may provide enough space in the .got section for a new entry, but this cannot be assumed in the general case. Therefore, this first approach has been completely discarded.

Since the forkserver is more complicated than the simple shared map update due to the presence of many controls on environment variables and communication with the fuzzer, the best approach is to inject the compiled forkserver code inside the target. This has been

done through a *Shared Object Injection*. In order to achieve a Shared Object Injection the *.dynamic* section of the target binary is modified. In particular :

- A *DT_NEEDED* entry is added with the name of the shared object to be injected.¹
- A *RUNPATH* entry is added containing the path in which is saved the actual shared object.

The shared object injected into the target binary, contains the code of the forking server that must be executed prior the main of the binary. In order to achieve the fact that a function of a shared object injected into a binary is executed before the main of the binary, that function must be declared as “constructor”. This means prepend to the function the string `__attribute__((constructor))`.

¹In reality the shared object name is a string that is written into the *.dynstr* section and into the *.dynamic* section is specified the offset of the string written into *.dynstr*

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List of Acronyms

BB Basic Block. 2, 4, 8–10, 13

CFG Control Flow Graph. 2, 4, 10

DBI Dynamic Binary Instrumentation. 3, 7–9, 12

IP Instrumentation Point. 4–6

IR Intermediate Representation. 6, 9

JIT Just in-time. 8

SBI Static Binary Instrumentation. 2, 7–9, 12

SHM Shared Memory. 10, 11, 13

TB Translation Block. 8

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